Chapter 12 exercises

Identify each quantity as an observed statistic, model expectation, or estimator. Check symmetry and positive semidefiniteness for covariance matrices.

CH12-E1

Sample data:

$$3, 5, 5, 9.$$

1. Compute the sample mean.
2. Compute sample variance with denominator $N - 1$.
3. Compute descriptive mean squared deviation with denominator N.
4. Explain the two uses.

CH12-E2

A discrete random variable has mass function

x	0	1	2
$p(x)$	0.2	0.5	0.3

1. Find its expectation.
2. Find $\mathbb{E}[X^2]$ and $\mathrm{Var}(X)$.
3. For $Y = 3X - 2$, find mean and variance.
4. Check $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

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CH12-E3

Let $X \sim U(0, L)$.

1. Give the density and CDF.
2. For $L = 4$, find $\mathbb{P}(1 < X < 3)$.
3. Derive mean and variance in terms of L, including $\mathrm{Var}(X) = L^2/12$, and interpret scaling.
4. Describe the moments after translating the interval to $[a, a + L]$.

CH12-E4

Joint probabilities:

$$P = \begin{bmatrix} 0.2 & 0.1 \\ 0.3 & 0.4 \end{bmatrix},$$

Chapter 12 exercises (continued)

Rows correspond to $X = 0, 1$, and columns to $Y = 0, 1$.

1. Compute $P(X = 0), P(X = 1), P(Y = 0), P(Y = 1)$.
2. Compute $\mathbb{E}[X], \mathbb{E}[Y], \mathbb{E}[XY]$.
3. Compute covariance and correlation.
4. Decide independence.

CH12-E5

A two-dimensional random vector takes the following three values, each with probability $1/3$.

$$X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad X_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad X_3 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

These subscripts index possible vector values, not three components of the random vector.

1. Find the mean vector.
2. Find the centered possible values.
3. Compute $V = \mathbb{E}[ZZ^T]$ from their outer products.
4. Determine whether it is positive definite or only semidefinite.

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CH12-E6

Let

$$X \sim N(\mu, V), \qquad \mu = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

$$V = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

1. Write the joint density.
2. Give the equation of a density level curve.
3. For $Y = (X_1, 3X_2)^T$, find mean and covariance.
4. Choose an SPD covariance with nonzero off-diagonal entry and write the resulting density.

CH12-E7

Measurements:

$$b_1 = x + u_1, \quad b_2 = x + u_2,$$

Chapter 12 exercises (continued)

Here x is a fixed unknown constant. The error vector $(u_1, u_2)^T$ has mean zero and covariance $V = \mathrm{diag}(\sigma_1^2, \sigma_2^2)$, with $\sigma_1, \sigma_2 > 0$. A diagonal covariance means uncorrelated errors; independence and normality are unnecessary.

1. Write the weighted objective.
2. Derive the parameter estimate.
3. Find its variance.
4. Evaluate the estimate for $\sigma_1 = 1, \sigma_2 = 1.5, b_1 = 10, b_2 = 11$.

CH12-E8

Model $b = Ax + E$, with E zero mean and known SPD covariance V, fixed design A. Let

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad V = \mathrm{diag}(1, 4, 9).$$

1. Write the weighted normal equations.
2. Compute $(A^T V^{-1} A)^{-1}$ symbolically.
3. State the BLUE assumptions and discuss whether normality is required.
4. Describe ways to handle a singular covariance estimate and what must be reported.

CH12-E9

1. Bound the rank of $\widehat{V}_{n-1} = \frac{1}{n-1} X_c X_c^T$, especially for $M \gg n$.
2. Let X have finite second moments, mean μ, and covariance V. Let independent $Z \sim N(0, W)$. Prove the sum has mean μ and covariance $V + W$. State when the sum is also Gaussian.
3. Interpret correlation and covariance rank for two perfectly linearly related variables, including constant-variable cases.
4. Explain why covariance does not identify a causal effect, using a common-cause example.

CH12-E10

Combine covariance spectral geometry, whitening, and WLS.

1. Let

$$V = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

Chapter 12 exercises (continued)

Verify SPD, find eigenvalues and orthonormal eigenvectors, and state the whitening scale in each direction. 2. Let $b = Ax + e$, where

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad V_e = \mathrm{diag}(1, 4, 9), \qquad b = (2, 5, 4)^T.$$

Form and solve the weighted normal equations, and find the weighted residual and $J(\widehat{x})$. 3. For the same A, b, incorrectly setting $V_e = I$, find the OLS estimate and its difference from WLS. 4. Give sufficient assumptions for BLUE. Which steps require normality?

CH12-E11

1. Let X be uniform on $-1, 0, 1$, $Y = X^2$. Find both means, variances, and covariance. Test independence using conditional probability.
2. For a sum S_N of N independent fair 0/1 variables, derive the values, spacing, and probabilities of standardized Z_N. Give the histogram heights suitable for comparison with normal density.
3. Independently draw $X_i \sim U(-1, 1)$, and estimate $1/3$ by $N^{-1} \sum_i X_i^2$. Find its mean and variance, and a sufficient N for Chebyshev to bound the probability of absolute error at least 0.01 by 0.05.
4. With IID observations and finite positive variance alone, does CLT guarantee any prescribed accuracy at $N = 30$? Distinguish a finite-sample bound from asymptotic convergence.

CH12-E12

Estimate one fixed unknown constant x, without state evolution or process noise. The old estimate is unbiased with error variance $W_0 = 1$, and its observed value is $\widehat{x}_0 = 10$. New observations satisfy $b_i = x + \eta_i$, $\mathbb{E}[\eta_i] = 0$, $R_1 = 9, R_2 = 1$, with observed values $b_1 = 13, b_2 = 9$. Assume η_1, η_2 are mutually uncorrelated and each is uncorrelated with the old estimation error $\widehat{x}_0 - x$. Both measurement matrices are $H = 1$.

Parts 1 and 3 also require the vector derivation. Treat the old estimate as an unbiased linear WLS summary of the same fixed state, with $W_0 \succ 0$. The new model has $H \in \mathbb{R}^{m \times n}$, $R = \mathrm{Cov}(\eta) \succ 0$, and new errors uncorrelated with the old estimation error. Normality is unnecessary.

Chapter 12 exercises (continued)

1. Derive $\widehat{x}_1 = \widehat{x}_0 + K(b_1 - H\widehat{x}_0)$ from the combined normal equations, and prove $K = W_0 H^T (R + HW_0 H^T)^{-1}$.
2. Compute the gains, estimates, and variances of both sequential updates, and compare with the batch solution.
3. Prove $W_1 \preceq W_0$. Why must a complete copy of an earlier observation not be counted again using the uncorrelated-error update?

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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